

COMS/SE 4170 SOFTWARE TESTING SPRING 2026 Combinatorial Testing

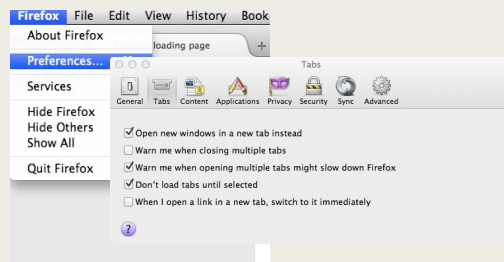
1

Configurability



2

Configurability



3

Configuration Interfaces

☐ Warn me when closing multiple tabs

4

Configuration Interfaces

☐ Warn me when closing multiple tabs

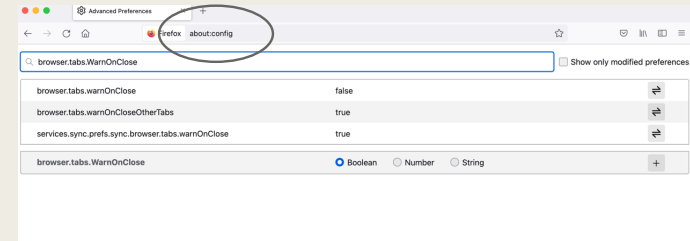
Prefs.js



```
user_pref("browser.tabs.warnOnClose", false);
```

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Configuration Interfaces



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Configuration Interfaces



Proceed with Caution

Changing advanced configuration preferences can impact Firefox performance or security.

☒ Warn me when I attempt to access these preferences

Accept the Risk and Continue

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Enter Query Sequence

Enter accession number(s), gi(e), or FASTA sequence(s)

gactacgatcgggc

BLAST Search database Nucleotide collection (nr/nt) using Megablast (Optimize for highly similar sequences)

Algorithm parameters

General Parameters

Max target sequences: 100

Short queries: Automatically adjust parameters for short input sequences

Expect threshold: 10

Word size: 28

Max matches in a query range: 0

Scoring Parameters

Match/Mismatch Scores: 1, -2

Gap Costs: Linear

Filters and Masking

Filter: Low complexity regions

Mask: Mask for lookup table only

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Enter Query Sequence

Enter accession number(s), g(s), or FASTA sequence(s) [?](#) Clear Query subrange [?](#)

From

To

BLAST Search database Nucleotide collection (nr/nt) using Megablast (Optimize for highly similar sequences) Show results in a new window

Algorithm parameters Restore default search parameters

General Parameters

Max target sequences: 100 [?](#)
Select the maximum number of aligned sequences to display [?](#)

Short queries: ☒ Automatically adjust parameters for short input sequences [?](#)

Expect threshold: 10 [?](#)

Word size: 28 [?](#)

Max matches in a query range: 0 [?](#)

Scoring Parameters

Match/Mismatch Scores: 1,-2 [?](#)

Gap Costs: Linear [?](#)

Filters and Masking

Filter: ☒ Low complexity regions [?](#)
☒ Species-specific repeats for: Homo sapiens (Human) [?](#)

Mask: ☒ Mask for lookup table only [?](#)
☐ Mask lower case letters [?](#)

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Enter Query Sequence

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Mask: ☒ Mask for lookup table only [?](#)
☐ Mask lower case letters [?](#)

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Firefox

Advanced Preferences

Search preference name Show only modified preferences

accessibility.AOM.enabled	false	?
accessibility.ARIAReflection.enabled	false	?
accessibility.accesskeycausesactivation	true	?
accessibility.blockautorefresh	false	?
accessibility.browsewithcaret	false	?
accessibility.browsewithcaret_shortcut.enabled	true	?
accessibility.cache.enabled	false	?
accessibility.force_disabled	0	?
accessibility.indicator.enabled	false	?
accessibility.monospeech.enabled	false	?
accessibility.mouse_focuses_formcontrol	2	?
accessibility.support.url	https://support.mozilla.org/%\$LOCALE%\$kb/accessibility-services	?
accessibility.tabfocus_applies_to_xul	true	?

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Challenge for Testing

Encoding	Format	Cache Level	Closed-Captioning	Network Access
MPEG	Audio	Low	Yes	Yes
RAW	Video	Medium	No	No
WAV	Stream	High		

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Media Player

Encoding	Format	Cache Level	Closed-Captioning	Network Access
MPEG	Audio	Low	Yes	Yes
RAW	Video	Medium	No	No
WAV	Stream	High		

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Media Player

Encoding	Format	Cache Level	Closed-Captioning	Network Access
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Media Player

Encoding	Format	Cache Level	Closed-Captioning	Network Access
MPEG	Audio	Low	Yes	Yes
RAW	Video	Medium	No	No
WAV	Stream	High		

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Testing the Player

Encoding	Format	Cache Level	Closed-Captioning	Network Access
MPEG	Audio	Low	Yes	Yes
RAW	Video	Medium	No	No
WAV	Stream	High		

Test Case: Open video and play to completion

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Interaction Fault

Encoding	Format	Cache Level	Closed-Captioning	Network Access
MPEG	Audio	Low	Yes	Yes
RAW	Video	Medium	No	No
WAV	Stream	High		

Test Case: Open video and play to completion



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Configuration-Dependent Security Bugs (CERT)

Overview
 Apache Struts 2 framework, versions 2.5 to 2.5.12, with REST plugin insecurely deserializes untrusted XML data. A remote, unauthenticated attacker can leverage this vulnerability to execute arbitrary code in the context of the Struts application.

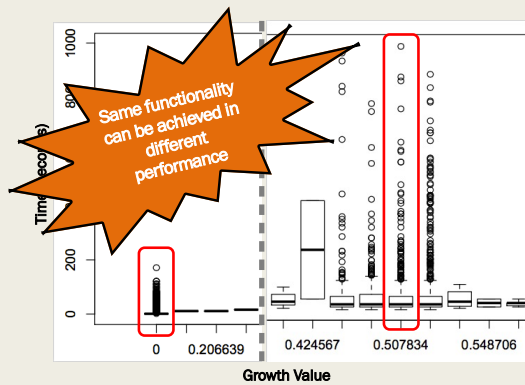
Description
 CWE-502: Deserialization of Untrusted Data - CVE-2017-6805
 In Apache Struts 2 framework, versions 2.5 to 2.5.12, the REST plugin uses `StrutsValueHandler` with an instance of `XStream` to deserialize XML data. Because there is no type filtering, a remote, unauthenticated attacker may send a specially crafted XML payload to execute arbitrary code in the context of the Struts application.
 Refer to the researcher's blog post for more information about this vulnerability. A Metasploit module with exploit code is publicly available.

Remove or limit the REST plugin
 If it is not used, consider removing the REST plugin. Per the vendor, it is also possible to limit its functionality to normal server pages or JSON with the following configuration change in `struts.xml`:

```
<constant name="struts.action.extension" value="html,json" />
```

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Also Impacts Program Performance



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Configuration Dependence

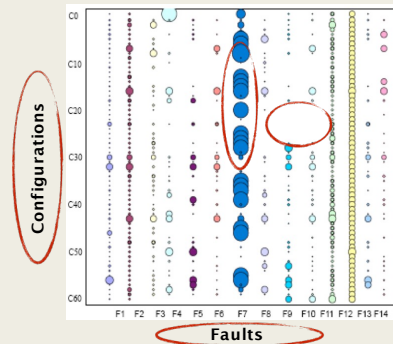


Figure credited to Xiao Qu

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Real Configuration Spaces

Encoding	Format	Caching Level	Closed-Captioning	Network Access
MPEG	Audio	Low	Yes	Yes
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WAV	Stream	High		

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Real Configuration Spaces

$$3 \times 3 \times 3 \times 2 \times 2 = 108 \text{ configurations}$$

Encoding	Format	Caching Level	Closed-Captioning	Network Access
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Real Configuration Spaces

$$3 \times 3 \times 3 \times 2 \times 2 = 108 \text{ configurations}$$

$$10 \text{ features with 5 options} = 9,765,625 \text{ configs}$$

Encoding	Format	Caching Level	Closed-Captioning	Network Access
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Real Configuration Spaces

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$$10 \text{ features with 5 options} = 9,765,625 \text{ configs}$$

$$4 \text{ hours to run test suite} \rightarrow 4,459 \text{ years to run}$$

Encoding	Format	Caching Level	Closed-Captioning	Network Access
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Real Configuration Spaces

$3 \times 3 \times 3 \times 2 \times 2 = 108$ configurations

10 features with 5 options = 9,765,625 configs

4 hours to run test suite $\rightarrow$ 4,459 years to run

gcc optimizer (199 options) $\rightarrow 10^{61}$

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Real Configuration Spaces

$3 \times 3 \times 3 \times 2 \times 2 = 108$ configurations

10 features with 5 options = 9,765,625 configs

4 hours to run test suite $\rightarrow$ 4,459 years to run

gcc optimizer (199 options) $\rightarrow 10^{61}$

Linux > 10,000 features



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Combinatorial Interaction Testing (CIT)

- Sample the space so that all **t-way** combinations of values occur **AT LEAST** once
- **t** is defined as strength of testing

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Combinatorial Interaction Testing

	Encoding	Format	Caching Level	Closed-Captioning	
1	MPEG	Stream	Medium	Yes	Yes
2	RAW	Video	High	No	No
3	MPEG	Video	Low	No	Yes
4	WAV	Stream	High	No	Yes
5	RAW	Stream	Low	Yes	No
6	MPEG	Audio	High	Yes	No
7	WAV	Video	Medium	Yes	No
8	RAW	Audio	Medium	No	Yes
9	WAV	Audio	Low	Yes	Yes

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Combinatorial Interaction Testing (CIT)

	Encoding	Format	Caching Level	Closed-Captioning	
1	MPEG	Stream	Medium	Yes	Yes
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3	MPEG	Video	Low	No	Yes
4	WAV	Stream	High	No	Yes
5	RAW	Stream	Low	Yes	No
6	MPEG	Audio	High	Yes	No
7	WAV	Video	Medium	Yes	No
8	RAW	Audio	Medium	No	Yes
9	WAV	Audio	Low	Yes	Yes

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Combinatorial Interaction Testing (CIT)

	Encoding	Format	Caching Level	Closed-Captioning	
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6	MPEG	Audio	High	Yes	No
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8	RAW	Audio	Medium	No	Yes
9	WAV	Audio	Low	Yes	Yes

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Combinatorial Interaction Testing (CIT)

	Encoding	Format	Caching Level	Closed-Captioning	
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7	WAV	Video	Medium	Yes	No
8	RAW	Audio	Medium	No	Yes
9	WAV	Audio	Low	Yes	Yes

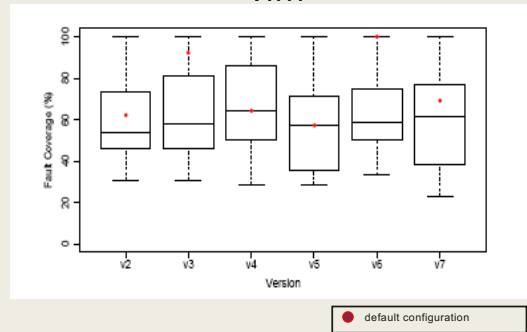
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Higher Strength CIT

- Strength (t) = 3 - all three-way combinations
- Needs *at least* 27 combinations

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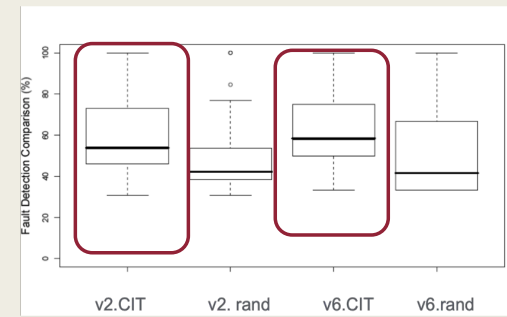
CIT vs. Default Configuration of vim



[X. Qu et al. ISSTA, 2008]

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Fault Detection

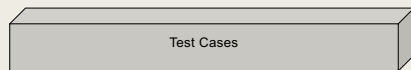


[X. Qu et al. ISSTA, 2008]

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CIT in Configurable Systems

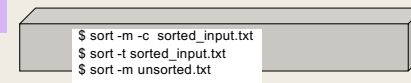
Traditional
CIT



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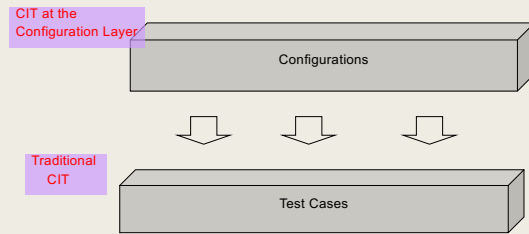
CIT in Configurable Systems

Traditional
CIT



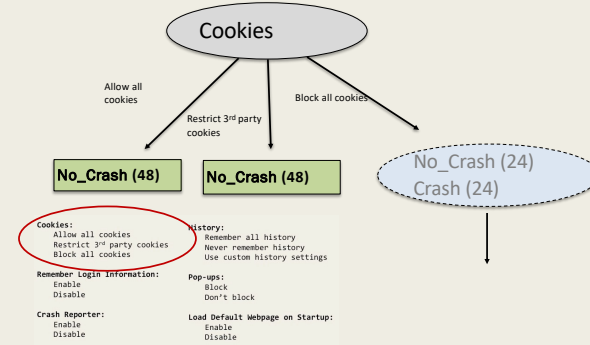
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CIT in Configurable Systems



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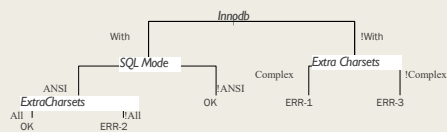
Classification: Decision Tree



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Characterizing Test Failures

- Use machine learning techniques (classification trees) to model option setting patterns that explain test failures



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